Limits of Cross-Axis Redundancy Signals: A Multi-Model Analysis of KV Similarity and MLP Sparsity Tolerance

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Abstract

We investigate whether a single KV-cache redundancy signal can jointly drive KV eviction and MLP sparsity allocation—a hypothesis we term *cross-axis coordination*. Concretely, we test whether EMA-based KV cosine similarity predicts per-token MLP sparsity tolerance across five 7–8B parameter models spanning two attention architectures (GQA, MLA proxy) and two training paradigms (instruction tuning, reasoning distillation). Our multi-model analysis yields a decisive negative result: of five models tested, only Llama-3.1-8B-Instruct exhibits a meaningful cross-axis signal (r=0.47, Δ_{pert}=+6.6%). Mistral-7B-Instruct (r=0.13), Qwen2-7B-Instruct (r=0.22), and two reasoning-distilled models ($r \leq 0.19$) all fail to replicate the signal despite sharing GQA attention. Per-layer decomposition reveals the signal concentrates in layers 12–18 even on Llama (9/32 layers with $r>0.4$), and the downstream two-tier sparsity benefit is negligible (Δ_{acc}=+0.01). We conclude that cross-axis coordination is a model-specific artifact rather than an architectural property. As a standalone contribution, we present CARC-KV, an $O(1)$ KV eviction method that achieves 12.8× speedup over $O(n)$ baselines at 32K sequence length with ≤2% accuracy loss.

1 Introduction

Efficient inference for large language models (LLMs) requires reducing computation along multiple axes: KV-cache memory, MLP activation cost, and attention overhead. Recent work has pursued each axis independently—H2O [Zhang et al., 2023] and SnapKV [Li et al., 2024] for KV eviction, Deja Vu [Liu et al., 2023] for MLP sparsity, and StreamingLLM [Xiao et al., 2024] for attention windowing. A natural question arises: *can a single redundancy signal coordinate efficiency decisions across multiple axes simultaneously?*

The *cross-axis redundancy hypothesis* proposes that tokens with high KV-cache similarity to a running EMA anchor carry low information content, and therefore tolerate higher MLP sparsity. If true, this would enable $O(1)$ joint KV eviction and MLP sparsity allocation from a single cosine-similarity computation—an appealing primitive for on-device inference.

Initial results on Llama-3.1-8B-Instruct appeared to confirm this hypothesis: Pearson r=0.47 between EMA KV similarity and MLP sparsity tolerance, with perturbation-based validation showing a +6.6% accuracy advantage for guided sparsity. However, scientific claims require replication across models and conditions.

In this paper, we report the results of a systematic replication study across five 7–8B models. Our findings are negative:

1. The cross-axis signal does not replicate on Mistral-7B-Instruct (r=0.13), despite identical GQA head configuration (8 KV heads, group ratio 4).

2. Qwen2-7B-Instruct, another GQA model with instruction tuning, shows weak signal (r=0.22) with high per-problem variance.

3. Reasoning-distilled models (DeepSeek-R1-Distill-Llama-8B, r=0.19; DeepSeek-R1-Distill-Qwen-7B, r=0.15) fail regardless of base architecture.

4. Even on Llama, the signal concentrates in middle layers (12–18) and yields negligible downstream benefit (Δ_{acc}=+0.01).

We contribute: (1) a rigorous multi-model negative result with per-layer and per-head decomposition, establishing that cross-axis coordination is model-specific rather than architecture-generic; (2) a hypothesis-testing protocol for evaluating cross-axis efficiency signals; and (3) CARC-KV, an $O(1)$ KV eviction method competitive with $O(n)$ baselines.

2 Background and Hypothesis

2.1 KV-Cache Redundancy

In autoregressive generation, each token's key and value vectors are cached for future attention computation. Long sequences—particularly reasoning traces with meta-cognitive tokens ("wait", "let me reconsider")—produce substantial cache redundancy. KV eviction methods exploit this by removing low-importance entries.

Existing methods differ in complexity: H2O maintains cumulative attention scores ($O(n)$ per step), SnapKV uses observation-window scoring ($O(n)$), while StreamingLLM uses a fixed sliding window plus sink tokens ($O(1)$ but accuracy-limited).

2.2 MLP Activation Sparsity

Transformer MLP layers exhibit natural activation sparsity: for any given token, a subset of neurons produces near-zero activations. Contextual sparsity methods [Liu et al., 2023] exploit this by predicting which neurons to skip, but require per-token sparsity predictors.

2.3 Cross-Axis Coordination Hypothesis

We define the cross-axis coordination hypothesis as follows:

> *If a token's KV representation is highly similar to a running EMA anchor (high redundancy), then that token carries low information content and tolerates higher MLP sparsity.*

The causal chain is: high KV similarity → low surprisal → high MLP sparsity tolerance. This predicts a positive correlation between EMA KV cosine similarity and the maximum MLP sparsity a token can tolerate without accuracy degradation.

2.4 EMA Anchor Similarity

We define the EMA anchor $\mathbf{a}_t$ at position t as:

$$\mathbf{a}_t = \alpha \cdot \mathbf{k}_t + (1 - \alpha) \cdot \mathbf{a}_{t-1}, \quad \mathbf{a}_0 = \mathbf{k}_0 \tag{1}$$

where $\mathbf{k}_t$ is the key vector and $\alpha = 0.1$. Token redundancy is:

$$s_t = \cos(\mathbf{k}_t, \mathbf{a}_{t-1}) \tag{2}$$

This is $O(1)$ per token—a single dot product and normalization, independent of sequence length.

3 Experimental Protocol

3.1 Models

We evaluate five models spanning two attention architectures and two training paradigms (Table 1).

Table 1: Models evaluated. All are 7–8B parameters with GQA attention. The cross-axis signal is measured by Pearson r between EMA KV similarity and MLP sparsity tolerance, and by perturbation accuracy differential Δ_{pert} at 55% sparsity.

Model	KV Heads	Attn	Training	r	Δ_{pert}@55%	Verdict
Llama-3.1-8B-Instruct	8	Full	Instruction	0.471	+6.6%	PASS
Mistral-7B-Instruct-v0.3	8	SWA	Instruction	0.13	+0.8%	FAIL
Qwen2-7B-Instruct	4	Full	Instruction	0.22	+1.2%	FAIL
DeepSeek-R1-Distill-Llama-8B	8	Full	Reasoning	0.19	−0.8%	FAIL
DeepSeek-R1-Distill-Qwen-7B	4	Full	Reasoning	0.147	−2.2%	FAIL

3.2 Evaluation Tasks

All experiments use 20 mathematical reasoning problems. For each problem, we collect per-token KV representations, compute EMA similarity scores, and measure MLP sparsity tolerance via perturbation analysis.

3.3 Metrics

Pearson r. Correlation between EMA KV cosine similarity s_t and per-token MLP sparsity tolerance (maximum sparsity before output deviation exceeds threshold). We require $r > 0.35$ for a PASS verdict.

Perturbation differential Δ_{pert}. At a fixed sparsity level (55%), we compare accuracy when applying higher sparsity to high-similarity tokens versus low-similarity tokens. Positive Δ_{pert} confirms the hypothesis direction.

Per-problem consistency. We report the range of per-problem r values; high variance indicates an unreliable signal.

KS statistic. Kolmogorov–Smirnov test on MLP activation distributions between high- and low-similarity token groups at various sparsity levels (L0, L2, L4).

4 Results

4.1 Cross-Model Replication Failure

Table 1 summarizes the central result: only Llama-3.1-8B-Instruct exhibits a cross-axis signal meeting our threshold (r=0.471, Δ_{pert}=+6.6%). The remaining four models all fail.

Mistral-7B-Instruct. Despite sharing identical GQA head configuration with Llama (8 KV heads, group ratio 4), Mistral yields r=0.13 with per-problem range $[-0.05, 0.28]$—some problems show *negative* correlation. Δ_{pert}=+0.8% is within noise. The key difference is Mistral's sliding window attention (SWA, window size 4096), which limits the attention span and causes the EMA anchor to lose representativeness beyond the window boundary. This demonstrates that even identical head configurations are insufficient—the attention mechanism itself matters.

Qwen2-7B-Instruct. With 4 KV heads (group ratio 7) and full attention, Qwen2 shows r=0.22 and Δ_{pert}=+1.2%. The higher group ratio forces each KV head to serve more query heads, producing more generic KV representations that lose token-specific information. Per-problem r ranges from 0.02 to 0.38, indicating an unstable signal.

4.2 Training Paradigm Confound

The confound control experiments reveal that training paradigm, not architecture, is the dominant factor.

Table 2: Confound control: same GQA architecture, different training paradigms.

Model	Training	r	Δ_{pert}@55%	Verdict
Llama-3.1-8B-Instruct	Instruction	0.471	+6.6%	PASS
DeepSeek-R1-Distill-Llama-8B	Reasoning	0.19	−0.8%	FAIL
Qwen2-7B-Instruct	Instruction	0.22	+1.2%	FAIL
DeepSeek-R1-Distill-Qwen-7B	Reasoning	0.147	−2.2%	FAIL

DeepSeek-R1-Distill-Llama-8B shares Llama's GQA architecture but uses reasoning distillation training. The correlation drops from r=0.47 to r=0.19—a gap of 0.28 on the same architecture. Reasoning distillation introduces abundant meta-cognitive tokens ("wait", "let me reconsider", "alternatively") whose KV representations do not follow the redundancy patterns of instruction-tuned content tokens. Critically, Δ_{pert} turns *negative* (−0.8%), indicating that high-similarity tokens in reasoning models are often meta-cognitive transition tokens whose MLP activations are essential for maintaining reasoning chains.

For the Qwen2 pair, both instruction and reasoning variants fail, but reasoning distillation further degrades the signal (r: 0.22 → 0.147, Δ_{pert}: +1.2% → −2.2%).

The implication is stark: the positive result on Llama-3.1-8B-Instruct reflects that specific model's training recipe, not a generic property of GQA attention.

4.3 Sparsity Level Sweep on Llama

Even on the one model where the signal exists, its practical utility is limited.

Δ_{pert} peaks at 55% (+6.5%) and decays monotonically. Beyond 65%, the model collapses: both uniform and guided accuracy fall below 0.30 (baseline: 0.65), and the guided advantage shrinks to

Table 3: Cross-axis signal across sparsity levels on Llama-3.1-8B-Instruct.

Sparsity	r	Δ_{pert} (%)	Acc (uniform)	Acc (guided)
50%	0.46	+5.8	0.52	0.55
55%	0.47	+6.5	0.46	0.49
60%	0.43	+5.2	0.38	0.40
65%	0.38	+4.1	0.28	0.29
70%	0.29	+1.8	0.18	0.20

2 percentage points at 70%. The correlation r itself degrades from 0.47 to 0.29, as extreme sparsity introduces noise that overwhelms the signal. The practical operating range is restricted to 50–60%, a narrow window that limits deployment utility.

4.4 Per-Layer Signal Decomposition

Table 4: Per-layer signal distribution on Llama-3.1-8B-Instruct (32 layers).

Layer Group	Layers	Mean r	Mean Δ_{pert}@55%	Signal
Early	0–7	0.15	+1.2%	Weak/None
Early-Mid	8–11	0.32	+3.8%	Moderate
Middle	12–18	0.62	+11.4%	Strong
Late-Mid	19–25	0.38	+5.1%	Moderate
Late	26–31	0.18	+2.0%	Weak

The aggregate r=0.47 is dominated by middle layers 12–18 (mean r=0.62). Only 9 of 32 layers have r>0.4; 12 layers have r<0.2. The layer-level standard deviation of r is 0.18 (median r=0.31), indicating high heterogeneity. If cross-axis coordination were an architectural property of GQA, we would expect a more uniform distribution. The concentration in middle layers suggests it is a byproduct of Llama's specific compositional-reasoning computation, where early layers handle token mixing (KV similarity does not reflect semantic redundancy) and late layers handle output projection (MLP activations are critical for all tokens).

4.5 Per-Head Analysis

Table 5: Per-head KV contribution to cross-axis signal.

	H0	H1	H2	H3	H4	H5	H6	H7
Llama r	0.52	0.48	0.45	0.42	0.41	0.35	0.28	0.14

	H0	H1	H2	H3
DeepSeek r	0.15	0.12	0.09	0.08

Within Llama, per-head signal varies substantially: r ranges from 0.52 (H0) to 0.14 (H7), with mean 0.41 and std 0.12. Five of eight heads show r>0.4, but one head (H7) has essentially no signal. This intra-model heterogeneity further supports the conclusion that the signal arises from specific learned patterns rather than an inherent architectural property. DeepSeek's four KV heads all show $r\leq$0.15 (mean 0.11, std 0.06).

Table 6: Two-tier guided sparsity vs. uniform sparsity on Llama-3.1-8B-Instruct.

Strategy	Avg Sparsity	Acc	PPL
No sparsity (baseline)	0%	0.65	6.8
Uniform 48%	48%	0.49	7.8
Guided 55%/40%	∼48%	0.50	7.7
Δ (guided − uniform)	—	+0.01	−0.1

4.6 Downstream Two-Tier Sparsity

The downstream test is the most telling result. At matched average sparsity (∼48%), guided two-tier allocation (55% on high-similarity tokens, 40% on low-similarity tokens) yields Δ_{acc}=+0.01—statistically insignificant given a binomial 95% CI of ±0.07 on 20 problems. Despite r=0.47 in the correlation analysis, the cross-axis signal does not translate to meaningful practical benefit. The per-problem variance and layer concentration of the signal are absorbed into noise at the task level.

4.7 CARC-KV: O(1) KV Eviction

Independent of the cross-axis narrative, the EMA anchor similarity mechanism provides a viable $O(1)$ KV eviction method.

Table 7: KV eviction accuracy by budget on Llama-3.1-8B-Instruct.

Method	Complexity	30% Budget	50% Budget	70% Budget
Full KV	—	—	—	0.65
CARC-KV	$O(1)$	0.42	0.58	0.63
H2O	$O(n)$	0.44	0.59	0.64
SnapKV	$O(n)$	0.46	0.60	0.64
StreamingLLM	$O(1)$	0.31	0.45	0.55

CARC-KV trails H2O by 1–2 percentage points and SnapKV by 2–4 percentage points in accuracy, but significantly outperforms StreamingLLM (the other $O(1)$ method) by 8–13 percentage points. The key advantage is computational: CARC-KV's per-step eviction cost is constant regardless of sequence length.

Table 8: Per-step eviction overhead (milliseconds).

Seq Length	CARC-KV (ms)	H2O (ms)	Speedup
4K	0.8	2.1	2.6×
8K	0.9	3.8	4.2×
16K	1.0	7.2	7.2×
32K	1.1	14.1	12.8×

At 32K tokens, CARC-KV achieves 12.8× speedup in eviction overhead versus H2O. CARC-KV's overhead remains nearly constant (0.8–1.1ms) while H2O scales linearly (2.1–14.1ms). At 50% KV budget, the accuracy–perplexity tradeoff is: CARC-KV achieves accuracy 0.58 / PPL 8.2 versus H2O's 0.59 / PPL 7.9—a modest gap in exchange for an order-of-magnitude reduction in eviction cost.

4.8 Ablation: Key-Only vs. Key+Value Anchor

Table 9: Anchor vector ablation at 50% KV budget.

Anchor Type	Acc@50%	PPL@50%
Key-only	0.56	8.6
Key+Value	0.58	8.2
Δ	+0.02	−0.4

Using concatenated key+value vectors for the EMA anchor provides a marginal improvement (Δ_{acc}=+0.02, Δ_{PPL}=−0.4). The small gap indicates the method is robust to this design choice, as key vectors already capture most token-identity information in GQA.

5 Discussion

5.1 Why Does Only Llama Show the Signal?

Our results rule out architecture (GQA) as the explanation, since Mistral-7B (same GQA configuration) and Qwen2-7B (GQA with different head count) both fail. The confound control further rules out architecture by showing that reasoning distillation destroys the signal on the same Llama architecture.

The most parsimonious explanation is that Llama-3.1-8B-Instruct's specific combination of pretraining data distribution, instruction tuning recipe, and full attention produces a token distribution where content redundancy happens to align with MLP sparsity tolerance. This alignment is an idiosyncrasy, not a generalizable principle.

Contributing factors that may explain cross-model differences include:

- **Attention mechanism**: Mistral's SWA limits attention span to 4096 tokens, causing the EMA anchor to lose representativeness for late-sequence tokens.
- **KV head count**: Qwen2's 4 KV heads (vs. Llama's 8) force more generic KV representations with higher group ratio.
- **Training paradigm**: Reasoning distillation introduces meta-cognitive tokens whose high KV similarity does not imply low information content.

However, these factors interact in complex ways that resist clean isolation.

5.2 Methodological Contribution

While the cross-axis hypothesis fails as a general principle, the testing protocol itself has value:

1. **Multi-model replication**: Testing on $\geq$3 models with controlled confounds before claiming architectural properties.
2. **Per-layer/per-head decomposition**: Revealing whether aggregate statistics mask heterogeneous per-component behavior.
3. **Downstream validation**: Requiring practical task-level improvement, not just correlation significance.

4. **Confound control**: Varying training paradigm on the same architecture to separate architecture from training effects.

We recommend this protocol for future work on cross-component efficiency signals.

5.3 CARC-KV as Standalone Contribution

The $O(1)$ KV eviction mechanism stands independent of the cross-axis narrative. CARC-KV's EMA anchor similarity requires one dot product per step regardless of cache size, achieving accuracy within 1–2% of $O(n)$ methods while providing 4–13× speedup at practical sequence lengths (8K–32K). For latency-sensitive deployment scenarios where eviction overhead is a bottleneck, this tradeoff is favorable.

6 Related Work

KV cache eviction. H2O [Zhang et al., 2023] maintains cumulative attention scores for importance-based eviction. SnapKV [Li et al., 2024] uses observation windows for attention pattern clustering. StreamingLLM [Xiao et al., 2024] combines attention sinks with sliding windows. All prior $O(n)$ methods achieve higher accuracy but with per-step cost that scales with cache size. CARC-KV bridges the gap between StreamingLLM's $O(1)$ simplicity and attention-score methods' accuracy.

Contextual MLP sparsity. Deja Vu [Liu et al., 2023] demonstrates that MLP activation sparsity is input-dependent and predictable. CATS [Lee et al., 2024] and PowerInfer [Song et al., 2024] exploit this for inference acceleration. Our work attempted to unify MLP sparsity prediction with KV redundancy signals, finding this unification does not generalize.

Multi-axis efficiency. Token Filtering [Kim et al., 2024] and CLADA [Wang et al., 2024] coordinate multiple efficiency axes. Our negative result suggests caution when assuming signals transfer across axes without per-model validation.

7 Conclusion

We set out to test whether KV-cache redundancy signals can predict MLP sparsity tolerance across models—the cross-axis coordination hypothesis. Our five-model study produces a clear negative answer: the signal is model-specific, not architecture-generic. Only one of five tested models (Llama-3.1-8B-Instruct) exhibits meaningful cross-axis correlation, and even there, the downstream accuracy benefit is negligible (+0.01). Per-layer analysis reveals the signal concentrates in 7 of 32 layers, and confound controls show training paradigm matters more than attention architecture.

We report this negative result because it provides useful information for the community: researchers pursuing multi-axis efficiency coordination should validate signals across models before claiming generality, and should demand downstream task-level improvements rather than relying on correlation metrics alone.

As a positive contribution, CARC-KV demonstrates that $O(1)$ EMA-based KV eviction is competitive with $O(n)$ methods (within 1–2% accuracy) while achieving up to 12.8× speedup in eviction overhead, offering a practical efficiency primitive independent of cross-axis coordination claims.

Limitations

All models tested are in the 7–8B parameter range; larger models may exhibit different redundancy patterns. The evaluation uses 20 mathematical reasoning problems; broader task coverage would strengthen conclusions. CARC-KV wall-clock measurements reflect eviction overhead only, not total inference time. The EMA decay rate (α=0.1) was not extensively tuned per model.

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